

Advanced maternal age at first delivery as a risk factor for long-term cardiovascular disease

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Aims

An increased number of women initiate motherhood at an advanced age. Advanced maternal age (AMA) at first delivery poses unique clinical challenges; however, limited literature exists regarding potential long-term implications. We investigated the association between AMA at first delivery and maternal long-term cardiovascular disease (CVD).

Methods and results

A population-based retrospective cohort study included all primiparous women at a tertiary medical centre (1991–2021). Participants were stratified by age (<30, 30–34, 35–40, and > 40 years). Kaplan–Meier analysis estimated cumulative incidence rates of morbidity, and a Cox regression model investigated the association between AMA and long-term maternal CVD while adjusting for confounding variables. The study included 47 046 primiparous women; 43 426 (92.3%) were <30 years, 2807 (6.0%) were aged 30–34 years, 658 (1.4%) were aged 35–40 years, and 155 (0.3%) were >40 years. Kaplan–Meier survival analysis revealed a significant increase in the cumulative incidence of CVD associated with AMA, particularly among aged >40 years (Log Rank: $P < 0.001$). Specifically, hypertension, ischaemic heart disease, central vascular events, and peripheral vascular disease were significantly more frequent in older age groups. Cox regression analysis indicated that AMA was independently associated with an increased risk of long-term CVD after adjusting for confounders [aHR = 1.81 (95% CI, 1.41–2.68); $P < 0.001$].

Conclusion

AMA at first delivery is associated with increased long-term risk of cardiovascular morbidity. These results suggest the importance of tailored monitoring and preventive strategies for women of AMA at first delivery to potentially mitigate future cardiovascular risks.

Lay summary

First delivery at advanced maternal age is associated with an increased long-term cardiovascular risk, independent of pregnancy complications.

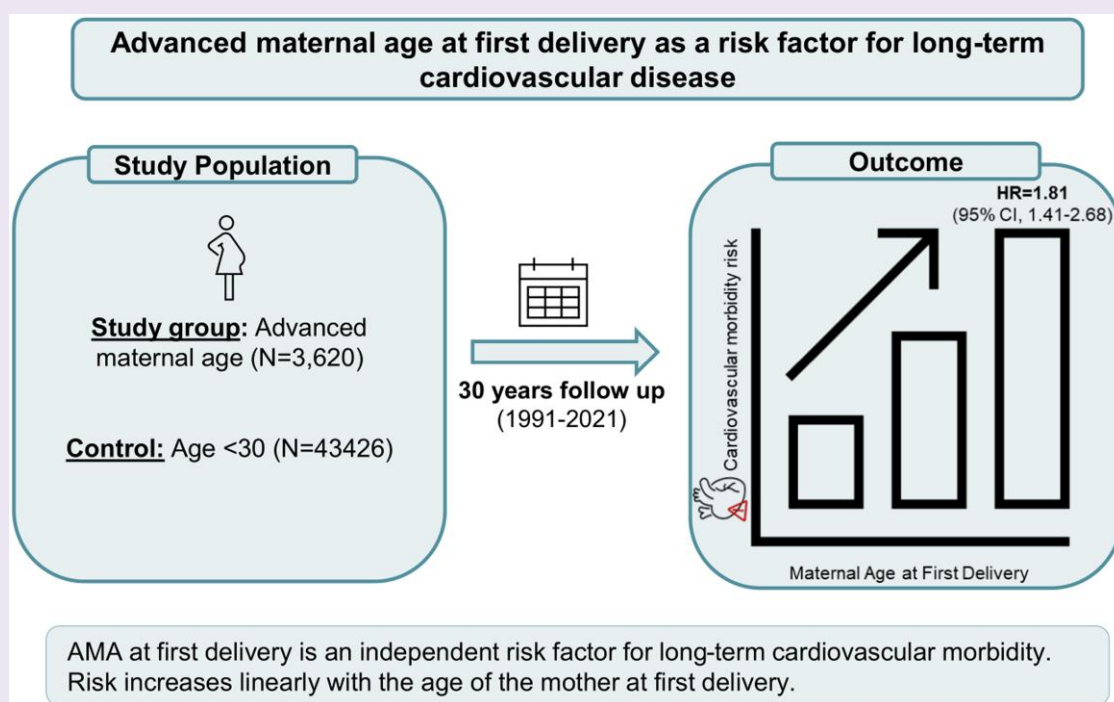
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Synopsis: First delivery at advanced maternal age increases long-term cardiovascular risk, independent of pregnancy complications.

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Graphical abstract



Schematic representation of the study design and main findings demonstrating the association between AMA at first delivery and higher long-term cardiovascular risk. HR, hazard ratio.

Keywords

Advanced maternal age • Cardiovascular • Developmental origins of health and disease • Pregnancy complications

Key Learning Points

What is already known:

- The age of first delivery is increasing globally, with well-documented short-term obstetric risks for advanced maternal age (AMA).
- While AMA is a recognized risk factor during pregnancy, its association with long-term cardiovascular morbidity remains controversial in current literature.

What this study adds:

- In this large-scale cohort of primipara, AMA at first delivery was independently associated with increased long-term cardiovascular morbidity, with the association most pronounced in women aged 35 and older.
- After adjusting for metabolic and obstetric confounders, maternal age ≥ 35 was associated with a five-fold higher incidence of hypertensive cardiac disease.
- Our findings demonstrate an increasing trend of cardiovascular incidence with advancing maternal age, underscoring the potential need for targeted long-term surveillance starting from the fourth decade of life.

Introduction

In recent years, there has been an increase in the number of women initiating motherhood at an advanced maternal age (AMA).¹ Women's prioritization of career advancement, coupled with advancements in assisted reproductive technology, are key contributing factors to this increasing trend.¹ While definitions vary, the most accepted cutoffs for AMA are age ≥ 35 years at the time of childbirth, age ≥ 40 years (classified as very AMA), and

age ≥ 45 years (classified as extremely advanced).¹ AMA poses a unique clinical challenge, due to its influence on pregnancy complications, outcomes, and maternal and foetal health.²

Cardiovascular diseases (CVD) remain a leading cause of mortality in men and women in all age groups.³ CVD encompasses a spectrum of conditions, including coronary heart disease, congestive heart failure (CHF), cerebrovascular disease, and peripheral arterial disease.⁴ The most common risk factors for CVD include hypertension, dyslipidaemia, Type 2 diabetes mellitus,

and gestational diabetes mellitus (GDM). These risk factors are known to contribute to atherosclerosis, increased cardiac workload, and endothelial dysfunction.^{4,5} Emerging evidence suggests that pregnancy-related complications serve as significant markers of future CVD.⁶ Specifically, complications such as pre-eclampsia and GDM are recognized as independent risk factors for long-term maternal atherosclerotic morbidity and cardiovascular complications.^{7,8}

AMA is known to be associated with an increased risk of several early pregnancy complications, including ectopic pregnancy, chromosomal abnormalities, and spontaneous abortion. In late pregnancy, it is linked to severe outcomes such as stillbirth, placental abruption, preterm delivery, amniotic fluid embolism, obstetric shock, as well as an increased incidence of GDM and hypertensive disorders.^{1,9,10} While these complications are generally considered to have short-term effects that typically resolve after childbirth, they may also have lasting implications for the long-term health of both the mother and the child.^{11,12} Consequently, extensive research has sought to explore the association between AMA and long-term adverse outcomes.^{9,13–18}

The relationship remains complex and is likely influenced by a combination of factors, leading to contradictory results in current literature. Some studies, such as those by Kahveci *et al.*, suggest that AMA increases glucose intolerance, possibly due to diminished insulin sensitivity, elevated levels of blood lipids, and elevated susceptibility to the development of Type 2 diabetes, as well as risks for myocardial infarction, CHF, and hypertensive disorders.^{12,19} Similarly, Khosla *et al.* found that hypertensive disorders during pregnancy are associated with a higher long-term risk of developing CVD.^{11,20,21} While these and other studies report a significant link,¹⁴ others suggest that the elevated susceptibility to CVD is primarily related to a history of obstetric complications,¹⁵ and some assert that AMA presents no heightened risk for CVD in subsequent life stages.¹³

Given this uncertainty and the rising prevalence of delayed childbearing, combined with the evidence of an increased incidence of pregnancy complications in primiparous women, understanding these relationships is crucial for informed clinical counselling of both patients and physicians. Hence, the aim of our study is to examine the association between AMA at first delivery and the long-term risk of maternal CVD.

Materials and methods

Study design and population

The study is a retrospective population-based cohort study that included all primigravida who delivered their firstborn between the years 1991 and 2021. The studied population was stratified into four age groups (<30 years old, 30–34 years old, 35–40 years old, and >40 years old). The primary outcome was defined as long-term maternal cardiovascular and vascular diseases of the mother, which was assessed using a predefined set of ICD-9 codes as recorded in medical institutions, including conditions such as ischaemic heart disease (IHD), hypertension (HTN), and peripheral vascular disease. Women with known CVD prior to the first pregnancy were excluded.

Setting

The research was conducted at Soroka University Medical Center (SUMC), the only tertiary hospital located in southern Israel. This facility provides medical care to a population exceeding one million.

Data collection method

The study is based on two computerized datasets: the first is that of perinatal births from the obstetric and gynaecologic department in SUMC, including information that was documented by obstetricians following delivery. The second dataset consists of hospitalization records from SUMC as well as Clalit HMO community services records, including demographic data and medical diagnoses, classified according to ICD-9 codes. Data were cross-checked, and the two databases were merged to create a comprehensive dataset. Diagnoses and relevant data were classified and coded using the International Classification of Diseases (ICD-9) system.

Statistical analysis

Statistical analyses were performed using SPSS software (version 29.0). Univariate analysis was conducted to assess differences in baseline characteristics across age groups. The groups were categorized based on maternal age at the time of first delivery. χ^2 tests were used to assess significant differences in the incidence of CVD and vascular morbidity between the groups.

Cumulative incidence rates were estimated using Kaplan–Meier analysis, which was used to facilitate the assessment of these rates over time, stratified by age group. For the survival analysis, time zero was defined as the date of the index delivery. Follow-up duration was calculated from time zero until the first occurrence of a cardiovascular event, death, or the end of the study period. Participants who reached the end of the follow-up period without experiencing an event or who died from non-cardiovascular causes were censored. While the study population was stratified into four age groups for descriptive purposes, the Kaplan–Meier survival analysis utilized a dichotomous cut-off of 35 years. This was done to align with the clinically accepted definition of AMA and to maintain sufficient statistical power for the comparison of cumulative incidence rates. Additionally, the log-rank test was utilized to compare survival curves between women who had their first delivery before age 35 and those who had their first delivery at age 35 or older in order to discern any significant disparities.

To assess the independent association between AMA and long-term maternal cardiovascular morbidity, multivariable Cox proportional hazards regression models were constructed. The models were adjusted for potential confounding variables selected based on their known clinical relevance as cardiovascular risk factors, as well as their statistical significance in the univariate analysis ($P < 0.05$). Those included maternal ethnicity, fertility treatments, and pregnancy-related complications: pre-gestational and gestational diabetes and pre-eclampsia (PET; defined by ICD-9 codes, see [Supplementary](#)). The primary multivariable model was adjusted for obesity, which is a key cardiovascular risk factor. However, in order to control other important variables, we have performed an additional sensitivity analysis. In this analysis, chronic hypertension was specifically incorporated as a covariate utilization (ICD-9 codes 642.0, 642.4, and 642.5) to further verify the robustness of our findings. This approach confirmed that the association between maternal age and cardiovascular morbidity remains highly significant ($P < 0.001$) regardless of whether obesity or hypertensive status is used as a covariate.

To assess the proportionality of hazards for the Cox regression model, the proportional hazards assumption was thoroughly evaluated using two methods. First, we performed a visual inspection of the Kaplan–Meier (KM) curves, which demonstrated consistent parallelism over time. To further validate

Table 1 Characteristics of women and pregnancy complications across the age group categories

Characteristic	Maternal age, years				P value
	< 30 N (43 426)	30–34 N (2807)	35–40 N (658)	> 40 N (155)	
Smoking <i>n</i> (%)	198 (0.5)	17 (0.6)	7 (1.1)	5 (3.2)	<0.001
Obesity <i>n</i> (%)	347 (0.8)	44 (1.6)	14 (2.1)	4 (2.6)	<0.001
Diabetes <i>n</i> (%)	1373 (3.2)	281 (10)	103 (15.7)	37 (23.9)	<0.001
Ethnicity <i>n</i> (%)	43 436	2807	658	155	<0.001
Bedouin <i>n</i> (%)	24 763 (57)	617 (22)	153 (23.3)	25 (16.1)	
Jewish <i>n</i> (%)	18 663 (43)	2190 (78)	505 (76.7)	130 (83.9)	
Fertility treatments <i>n</i> (%)	1772 (3.6)	516 (18.4)	228 (34.7)	91 (58.7)	<0.001
Hypertensive disorder ^a <i>n</i> (%)	2681 (6.2)	224 (8)	69 (10.5)	30 (19.4)	<0.001
Preterm birth <i>n</i> (%)	4401 (10.1)	333 (11.9)	107 (16.3)	40 (25.8)	<0.001

P-values were calculated using χ^2 tests for categorical variables (expressed as *n*, %) and one-way ANOVA for continuous variables (expressed as mean *n*, %). *N* represents the total number of women in each age group.

^aHypertensive disorder—mild-server pre-eclampsia or eclampsia.

this statistically, we included interaction terms between the age variables and time within the model. These interaction terms were found to be non-significant, indicating that the proportional hazards assumption was met and that our model is appropriately specified.

Results

The study included 47 046 primipara, which were categorized into four age groups: under 30 (*n* = 43 426, 92.3%), 30–34 years (*n* = 2807, 5.96%), 35–40 years (*n* = 658, 1.39%), and age above 40 (*n* = 155, 0.32%). The median follow-up period was 11.2 years [standard deviation (SD) $\pm$ 7.7 years]. Given the relatively small sample size of the >40 age group, findings related to this specific cohort should be interpreted with caution.

Table 1 summarizes the characteristics of the patients across these age groups. Obesity was significantly more frequent in the older cohorts, specifically in women aged 35–40 (2.1%) and those aged >40 (2.6%), compared to younger cohorts (*P* < 0.001). Diabetes rates increased across age groups: under 30 years (3.2%); 30–34 years (10%), 35–40 years (15.7%), and age >40 years (23.9%) (*P* < 0.001). A difference was observed in rates of parturition between Jewish and Bedouin populations, with a higher proportion of Jewish women among those giving birth at ages 30 and above (30–34 years: 78%; 35–40 years: 76.7%; and >40 years: 83.6%). Regarding fertility treatments, an increasing rate was observed with age, reaching 58.7% in the age 40 and above.

Pregnancy complications across the different age groups are also present in Table 1. The rates of mild to severe pre-eclampsia or eclampsia were significantly higher in women aged 35 and above (12.17%, *P* < 0.001). Women in the age group of 35–40 and age 40 and above had a significantly higher rate of preterm birth (35–40 years: 16.3%; >40 years: 25.8%; *P* < 0.001). No statistically significant difference was observed in the rates of small for gestational age and intrauterine growth restriction between the different maternal age groups.

Table 2 presents long-term cardiovascular morbidity stratified by maternal age. Hypertension incidence increased with age

(age <30: 4.4%; age 30–34: 7.2%; age 35–40: 14.1%, age >40: 15.5%; *P* < 0.001). Similarly, higher rates of IHD, central vascular events, and peripheral vascular disease were observed in the older age group. No statistically significant differences were found in the incidence of structural valvular disease, arrhythmia, rheumatic heart disease, pulmonary heart disease, pericarditis, endocarditis, and myocarditis between the different maternal age groups.

Figures 1 and 2 present Kaplan–Meier survival analysis illustrating the cumulative incidence of total vascular and cardiovascular disease (respectively) in the different maternal age groups. The analysis demonstrates a higher cumulative incidence of both cardiovascular events and vascular disease associated with advancing age (Log Rank, *P* < 0.001).

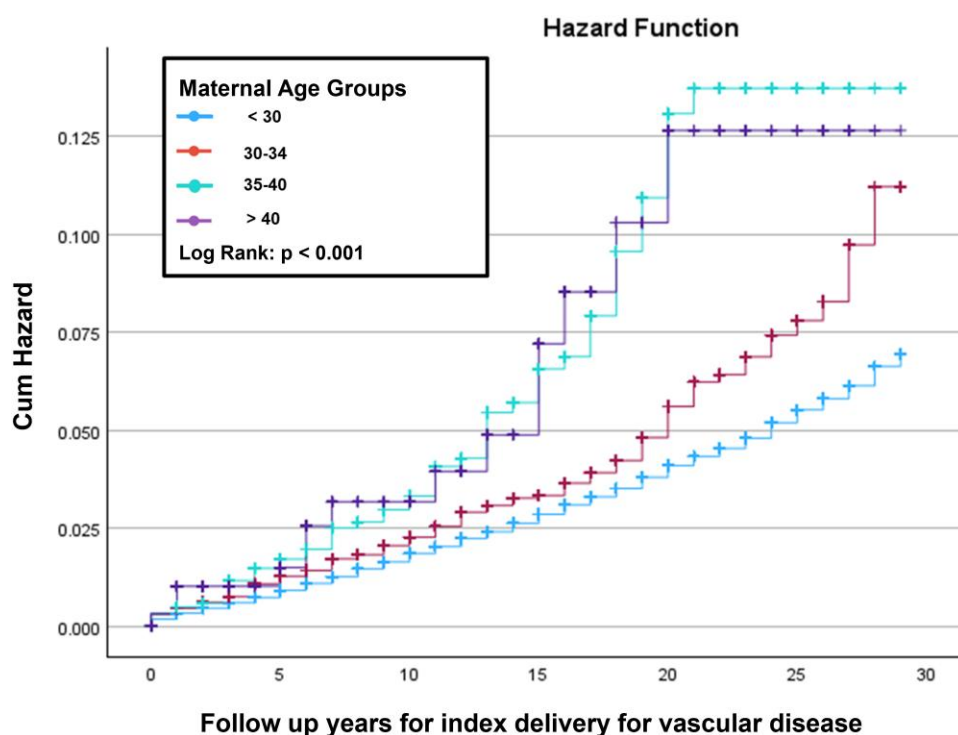
Table 3 presents the Cox regression model that was used to estimate the adjusted hazard ratio and 95% confidence interval (CI) for long-term CVD outcomes. After adjusting for potential confounders—including fertility treatments, diabetes (pre-gestational and GDM), PET, obesity, and ethnicity—AMA at first delivery remained significantly associated with an increased risk for cardiovascular morbidity. Compared to the <30 group, the adjusted hazard ratio was 1.304 in the 30–34 years group and 2.057 for women aged $\geq$ 35 years. The full list of variables included in the multivariable adjustment is detailed in the footnote of Table 3. Variables were selected based on clinical relevance.

To further validate our findings and the binary age categorization, a sensitivity analysis was performed comparing women aged <35 to those $\geq$ 35 at first delivery. Kaplan–Meier survival analysis demonstrated a significantly higher cumulative incidence of CVD in the age $\geq$ 35 group (χ^2 = 196.147; *P* < 0.001). In the multivariate Cox regression models, which adjusted for ethnicity, chronic hypertension, pre-eclampsia, diabetes, and fertility treatments, maternal age $\geq$ 35 was associated with an increased risk for long-term cardiovascular complications. However, this association varied across specific outcomes. While an upward but non-significant trend was observed for IHD, [aHR = 2.010 (95% CI, 0.700–5.776); *P* = 0.195], a statistically significant five-fold higher incidence was observed for

Table 2 Long-term maternal cardiovascular morbidity across the age group categories

Characteristic	Maternal age, years				P value
	< 30 N (43 426)	30–34 N (2807)	35–40 N (658)	> 40 N (155)	
Total cardiovascular disease <i>n</i> (%)	5910 (13.6)	405 (14.4)	140 (21.3)	39 (25.2)	< 0.001
Cardiac heart disease not specified <i>n</i> (%)	114 (0.3)	12 (0.4)	5 (0.8)	2 (1.3)	< 0.003
Hypertension <i>n</i> (%)	1890 (4.4)	202 (7.2)	93 (14.1)	24 (15.5)	< 0.001
Ischaemic heart disease <i>n</i> (%)	96 (0.2)	13 (0.5)	5 (0.8)	2 (1.3)	< 0.001
Central vascular event <i>n</i> (%)	292 (0.4)	44 (0.9)	25 (2)	5 (1.6)	< 0.001
Peripheral vascular <i>n</i> (%)	70 (0.1)	7 (0.1)	3 (0.2)	4 (1.3)	< 0.003
Embolism thrombosis <i>n</i> (%)	572 (0.7)	33 (0.6)	12 (1)	1 (0.3)	< 0.001
Structural valvular disease <i>n</i> (%)	149 (0.3)	10 (0.4)	3 (0.5)	1 (0.6)	0.085
Arrhythmia <i>n</i> (%)	4365 (10.1)	234 (8.3)	69 (10.5)	19 (12.3)	0.021
Rheumatic heart disease <i>n</i> (%)	49 (0.1)	2 (0.1)	1 (0.2)	0 (0)	0.876
Pulmonary heart disease <i>n</i> (%)	73 (0.2)	3 (0.1)	4 (0.6)	1 (0.6)	0.018
Pericarditis, endocarditis and Myocarditis <i>n</i> (%)	63 (0.1)	7 (0.2)	2 (0.3)	0 (0)	0.376

P-values were calculated using χ^2 tests to compare the incidence of cardiovascular morbidity across age groups. N represents the total number of women in each age group.

**Figure 1** Kaplan–Meier analysis illustrating the cumulative incidence of vascular disease in different maternal age groups.

hypertensive cardiac disease [aHR = 5.119 (95% CI, 3.555–7.371); $P < 0.001$]. The findings regarding IHD should be interpreted with caution due to the limited number of events and the resulting wide CI. Furthermore, an additional sensitivity model was performed incorporating chronic and

pregnancy-related hypertension as covariates (ICD-9 codes: 642.0, 642.4, 642.5). In this analysis, AMA remained a significant independent factor for long-term cardiovascular morbidity ($P < 0.001$), confirming the robustness of the primary findings.

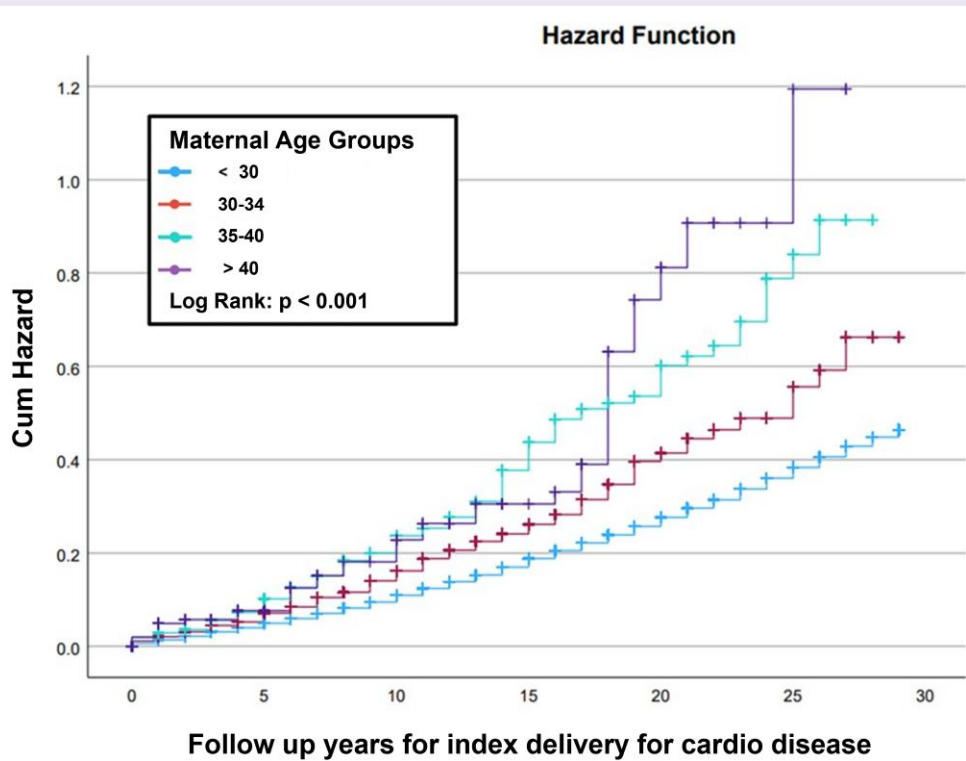


Figure 2 Kaplan–Meier analysis illustrating the cumulative incidence of total cardiovascular disease in different maternal age groups.

Table 3 Association between maternal characteristics and total long-term maternal cardiovascular and vascular morbidity; results of Cox regression analysis

Characteristic	HR (95% CL)	P value
Maternal age < 30	1.00	
Maternal age 30–34 years vs. < 30	1.304 (1.209–1.488)	< 0.001
Maternal age 35–40 years vs. < 30	2.057 (1.605–2.263)	< 0.001
Maternal age > 40 vs. < 30	1.810 (1.414–2.684)	< 0.001

HR, hazard risk; CI, confidence interval.
Mothers age < 30 years served as the reference group (HR 1.00).
The model was adjusted for maternal ethnicity, obesity (ICD-9 codes 649.11, 646.11), fertility treatments, pre-eclampsia (PET), and diabetes mellitus (including pre-gestational and gestational diabetes: ICD-9 codes 648.81, 648.01, 250.00, 250.01, 250.51, and 362.01).
P values were derived from the multivariable Cox proportional hazards model.
A separate sensitivity analysis was performed, adjusting for chronic hypertension instead of obesity, yielding consistent results (see Statistical Analysis section).

Discussion

Main findings

In the current study, we observed an increased incidence of long-term cardiovascular morbidity—specifically hypertension and IHD—in women with AMA at first delivery, emphasizing the complex relationship between pregnancy-related physiological changes at an advanced age and long-term cardiovascular health. Our results reveal not only a significant association but also a linear increase in morbidity rates across older age groups,

suggesting that maternal age is independently associated with increased cardiovascular morbidity even when controlling for other pregnancy complications.

Notably, our primary findings remained consistent when maternal age was examined as a binary variable (<35 vs. ≥35 years). This sensitivity analysis indicated that even after adjusting for fertility treatments and other clinical confounders, the incidence of cardiac complications was consistently observed in the older group.

Our findings align with the growing body of evidence emphasizing the importance of sex-based differences in

cardiovascular risk profiles. Recent studies have highlighted that reproductive history and sex-specific factors play a crucial role in determining long-term cardiovascular outcomes, particularly noting that hypertension may carry a disproportionately higher burden of risk in women.²² By observing a strong association between AMA and hypertensive cardiac disease, our study further supports the need for a nuanced, sex-aware approach to cardiovascular risk assessment in women.

Findings in the context of existing literature

Our findings challenge previous studies that either suggested no association between AMA at first delivery and long-term risk for CVD¹² or attributed this susceptibility to intervening pregnancy complications.¹⁴ Our data indicate that maternal age itself serves as an independent marker. This suggests that a first pregnancy at an advanced age acts as a natural physiological cardiovascular 'stress test', heightening physiological demand and potentially unmasking latent vulnerabilities regardless of the immediate obstetric outcome. Consequently, AMA should be recognized as a distinct clinical marker for long-term cardiovascular risk—specifically for hypertension and IHD—necessitating a proactive management approach.

Strengths and limitations

Our study's strengths include a large sample size and the unique context of a sole regional tertiary medical centre. This unique position, combined with integrated Clalit HMO records, enables near-complete capture of long-term diagnoses and follow-up for our data. Furthermore, the consistent data entry by physicians, complemented by medical secretary review, minimizes misclassification risk. Finally, our research questions were specifically tailored to align with the available dataset, further ensuring data integrity.

Despite its strengths, the retrospective design introduces inherent limitations, primarily the reliance on administrative coding, which may introduce errors. Furthermore, the lack of longitudinal data on postpartum factors—such as smoking status, BMI changes, lipid profiles, and blood pressure trajectories, as well as socioeconomic status—means the possibility of residual confounding cannot be excluded.

Over the 30-year study period, changes occurred in obstetric practice and cardiovascular prevention and diagnosis methods. To overcome these temporal challenges, we adjusted our final multivariate model for year of birth to account for secular trends, and the association between AMA and cardiovascular risk remained significant. Additionally, certain pregnancy complications (e.g. preterm birth or IUGR) were excluded from the multivariable model due to their established roles as major independent risk factors.

Regarding external validity, our single-centre cohort with a unique population structure—including distinct Jewish and Bedouin subgroups—strengthens internal consistency but may limit its applicability to different demographic patterns. Furthermore, the relatively small sample size of the >40 years group ($n = 155$, representing 0.3% of the total cohort) limits the precision of risk estimates for this subgroup and may result in wider CIs; therefore, the Hazard Ratios for women delivering at age >40 should be interpreted with caution. Finally, as an observational study, we can identify associations but cannot establish direct causality.

Clinical implications and postpartum surveillance

Given the global rise in maternal age at first pregnancy, our findings have significant clinical implications. While previous evidence regarding AMA and long-term cardiovascular risk has been inconsistent, this study clarifies that AMA at first delivery is an independent marker for CVD morbidity. This recognition necessitates tailored clinical monitoring and preventive strategies—both during and after pregnancy—to mitigate future cardiovascular risk.

Specifically, preventive strategies should be broadened to include earlier and more frequent screening for hypertension and IHD, the specific risks identified in our study. Effective management also requires improved co-ordination between obstetricians and primary care physicians to facilitate a seamless transition from the postpartum period to long-term care, supporting these women to remain within the healthcare framework after delivery. By integrating AMA into standard risk-stratification models, clinicians can shift from reactive treatment of established disease to proactive prevention, potentially mitigating the long-term cardiovascular burden in this growing population.

Conclusions

Previous studies have established that AMA at delivery is associated with an elevated risk of cardiovascular and vascular morbidity, largely attributed to complications arising during pregnancy. In our study, we observed that AMA at first delivery is independently associated with cardiovascular morbidity, even after adjusting for pregnancy-related complications and maternal characteristics. Moreover, while controlling these confounding variables, AMA at first delivery consistently remained significantly associated with long-term cardiovascular outcomes. We hypothesize that the pathophysiological mechanisms underlying this association are multifactorial, potentially related to challenges in physiological adaptation to pregnancy-induced changes. These findings underscore the necessity for further research to elucidate the mechanisms underlying the association between AMA and CVD, with the aim of developing targeted interventions and preventive strategies to enhance maternal health outcomes, particularly in higher age groups.

Supplementary material

Supplementary material is available at *European Heart Journal—Quality of Care and Clinical Outcomes* online.

Author contributions

Michal Yanoshvsky: Conceptualization, Methodology, Writing—original draft. Tamar Wainstock: Data curation, Formal analysis. Alex Rabinovich: Methodology, Supervision. Hagar Bami: Writing—review & editing. Ruslan Sergienko: Data curation, Formal analysis. Eyal Sheiner: Writing—review & editing. Roy Kessous: Conceptualization, Data curation, Formal analysis, Writing—review & editing, Supervision.

Michal Yanoshvsky (Conceptualization, Methodology [equal], Investigation [supporting], Writing—original draft [lead]), Tamar Wainstock (Data curation [supporting], Formal analysis, Software, Visualization [lead]), Alex Rabinovich

[Conceptualization, Methodology, Supervision, Writing—review & editing (supporting)], Hagar Bami [Validation, Writing—review & editing (supporting)], Ruslan Sergienko [Data curation, Funding acquisition, Software (supporting)], Eyal Sheiner [Validation, Writing—review & editing (equal)], and Roy Kessous [Conceptualization, Data curation, Methodology, Project administration, Supervision, Writing—review & editing (lead)]

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Conflict of interest: None declared. All authors have completed the ICMJE uniform disclosure form.

Data availability

The data underlying this article cannot be shared publicly due to the privacy of individuals that participated in the study. The data will be shared on reasonable request to the corresponding author.

Additional statements

* Preliminary results from this study were presented as an abstract at the 45th Annual Pregnancy Meeting (SMFM) in Denver, Colorado (2025).

The full manuscript has not been published and is not under consideration for publication elsewhere.

All authors have read and approved the final manuscript

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